Rising House Prices and Growing Gap

Problem Statement

Housing affordability in England and Wales has come under increasing pressure, but the change has not been the same everywhere. Between 2020 and 2024, house prices were affected by the pandemic and temporary tax changes, while household incomes grew at a different pace. This project combines HM Land Registry Price Paid Data with ONS gross disposable household income (GDHI) per head to measure housing affordability across local authorities from 2020 to 2024. It aims to identify where and when affordability deteriorated, how it varies by property type, and which factors are associated with local house prices.

Source Code: https://github.com/tasmiahshawrin/Rising-House-Prices-and-Affordability-Gap

Main Research Question

How did housing affordability, measured as median house price relative to GDHI per head, change across local authorities in England and Wales between 2019 and 2024, and how well do local income, property type and location explain differences in house prices?

Specific Research Questions:

1. How did median house prices and GDHI per head each change between 2020 and 2024, and which grew faster?
2. Which local authorities and regions are the least and most affordable, and did the gap between them widen?
3. How strongly is local income associated with median house price, and did this relationship change over the period?
4. Does affordability differ by property type (detached, semi-detached, terraced, flat)?
5. How did prices respond around 2020 to 2021 compared with 2022 to 2024?

Dataset Selection

1. **HM Land Registry Price Paid Data (Price Paid Data yearly file - GOV.UK):** Covers 5,051,767 residential property transactions (2020-2024), including transaction ID (tid), sale price, date, property type, new-build status, tenure, district, and county.
2. **ONS Gross Disposable Household Income (GDHI) per Head (Gross disposable household income per head - ONS):** Official annual time-series from 1997 to 2024, filtered to 2020–2024 across 318 local authorities. It captures total household income after taxes, national insurance, and social benefits, divided by population.

Ethical & Legal Compliance: Both datasets are licensed under the Open Government Licence v3.0 (OGL). To protect privacy, house numbers (PAON, SAON) and street names were removed during cleaning in line with GDPR and the Data Protection Act 2018. No data was fabricated or synthetically generated.

Data Validation and Processing

1. **House Price Dataset Preparation (HM Land Registry):** Five datasets from the HM Land Registry Price Paid Data, covering the years 2020-2024 were selected and combined into a single dataset. The following preprocessing steps were performed:

    - Data Cleaning: Whitespace was removed from all string columns to maintain consistency.
    - Data Type Standardization: The ‘price’ column was converted to float, while transaction dates were parsed into datetime format.
    - Missing Value Handling: The ‘postcode’, ‘saon’, ‘street’, and ‘locality’ columns contained substantial missing values and were therefore dropped.
    - Dataset Export: The resulting cleaned dataset was saved as cleaned_house_price_dataset.csv.

2. **Household Income Dataset Preparation:** The second dataset was obtained from the Office for National Statistics (ONS), containing Gross Disposable Household Income (GDHI) per head. The dataset was filtered to retain 318 geographical areas covering the period from 2020 to 2024. Following the selection and cleaning process, the resulting dataset was exported as cleaned_income_dataset.csv.
3. **Geographical Harmonization and Dataset Merging:** The initial merge using district names left 354,284 house price transactions unmatched. Naming inconsistencies, such as 'CITY OF BRISTOL' and 'BRISTOL, CITY OF', also caused mismatches. To address this, a geographical harmonization dictionary was created to standardize administrative names.

   After harmonization, both datasets were merged, creating a unified dataset for analyzing house prices, income, and affordability across geographical areas from 2020 to 2024.

Exploratory Data Analysis (EDA) and Summary Statistics

Transaction prices show strong positive skewness, mainly due to expensive properties in Central London. The mean price is higher than the median, indicating the influence of extreme values. Since the mean is higher than the median due to the influence of expensive luxury properties, using it could give a misleading representation of typical house prices and local purchasing power. Therefore, this study uses median house prices relative to GDHI per head to provide a more reliable measure of typical housing affordability: Affordability Ratio = Median House Price / GDHI per Head

	Metric	Transaction Price (£)	GDHI Per Head (£)	Affordability Ratio
0	Count	5,051,767	1,590	1,590
1	Mean	£397404	£24184	12.33x
2	Std Dev	£1594294	£10403	2.91x
3	Min	£1	£13909	4.9x
4	25% (Q1)	£173000	£19372	10.11x
5	50% (Median)	£270000	£22180	12.32x
6	75% (Q3)	£417500	£26161	14.37x
7	Max	£900000000	£171569	21.44x
8	Skewness	124.98	7.24	0.17

Research Question Analysis

1. How did median house prices and GDHI per head each change between 2020 and 2024, and which grew faster?

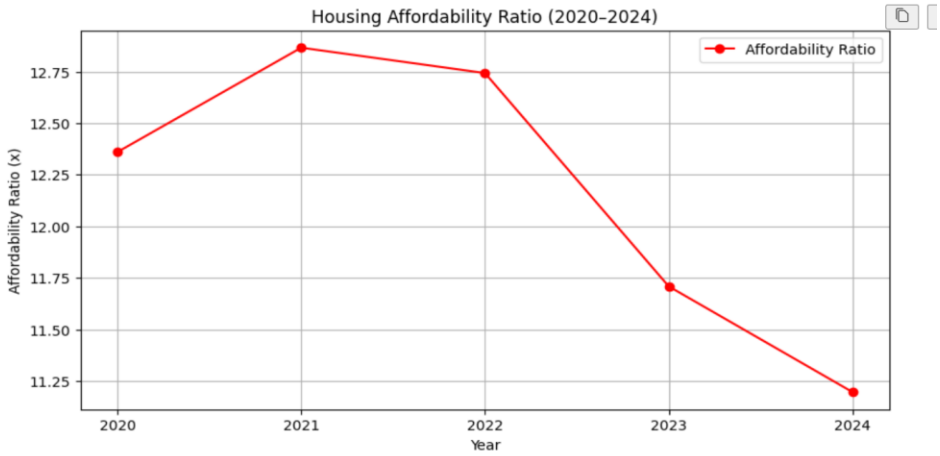
   Between 2020 and 2024, median GDHI per head increased by 23.64% (£20,226 to £25,008), while median house prices rose by 12% (£249,995 to £280,000). This shows that income grew almost twice as fast as house prices.

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Annual Summary (2020-2024)
   year  median_price  price_growth_%  median_income  income_growth_%  \
0  2020      249995.0             NaN        20226.0              NaN
1  2021      269995.0        8.000160        20985.5         3.755068
2  2022      280000.0        3.705624        21973.5         4.708013
3  2023      275000.0       -1.785714        23486.5         6.885567
4  2024      280000.0        1.818182        25007.5         6.476061

   affordability_ratio
0            12.360081
1            12.865788
2            12.742622
3            11.708854
4            11.196641
Cumulative Growth (2020 to 2024)
Median House Price: £249995 -> £280000 | Total Growth: 12.0%
Median GDHI/Head: £20226 -> £25008 | Total Growth: 23.64%
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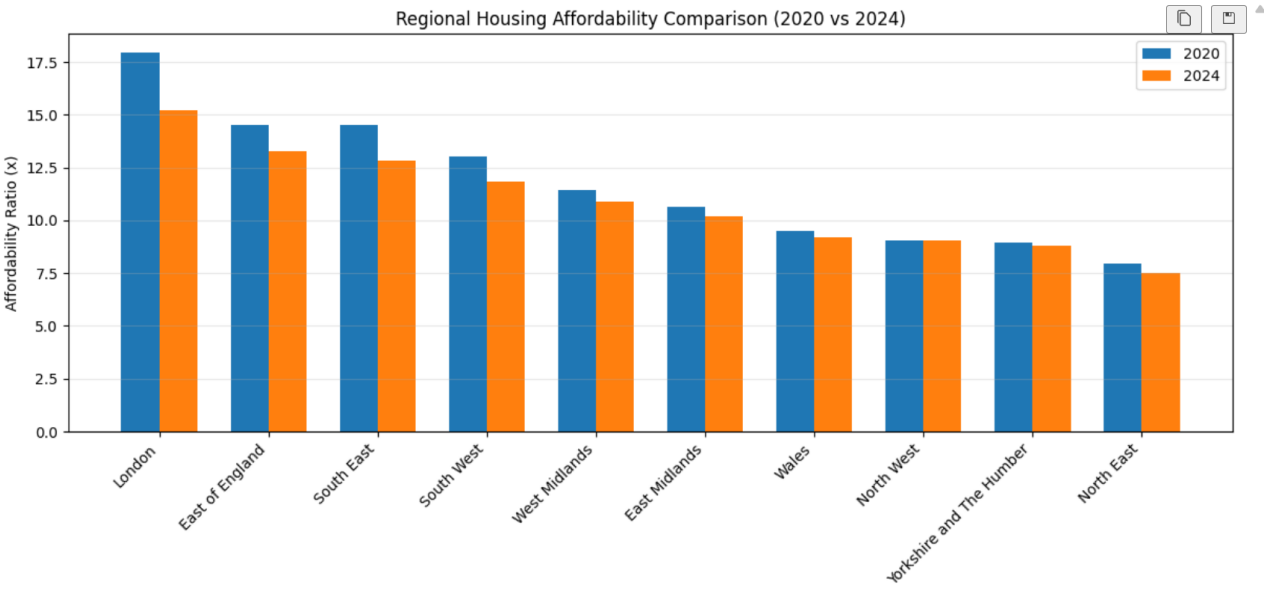
However, growth varied across the period with two distinct phases:

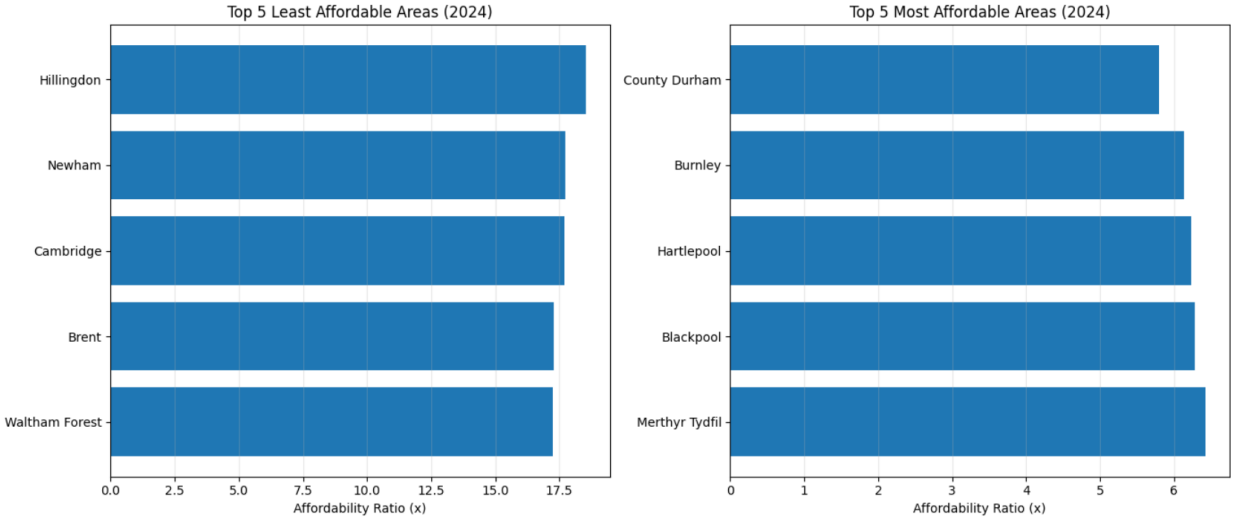
- 2020 to 2021: the ratio rose to about 12.9, so housing became less affordable.
- 2022 to 2024: the ratio fell sharply to about 11.2, so housing became more affordable.



2. Which local authorities and regions are least and most affordable, and did the gap between them widen?

- Regional Ranking: London is the least affordable region, followed by the East of England and South East. The North East is the most affordable, followed by Yorkshire and The Humber, the North West, and Wales.
- Local Authority Ranking: In 2024, the least affordable are Hillingdon, Newham, Cambridge, Brent, and Waltham Forest. The most affordable are County Durham, Burnley, Hartlepool, and Blackpool. The affordability gap between the least and most affordable local authorities decreased.





3. How strongly is local income associated with median house price, and did this relationship change over the period?

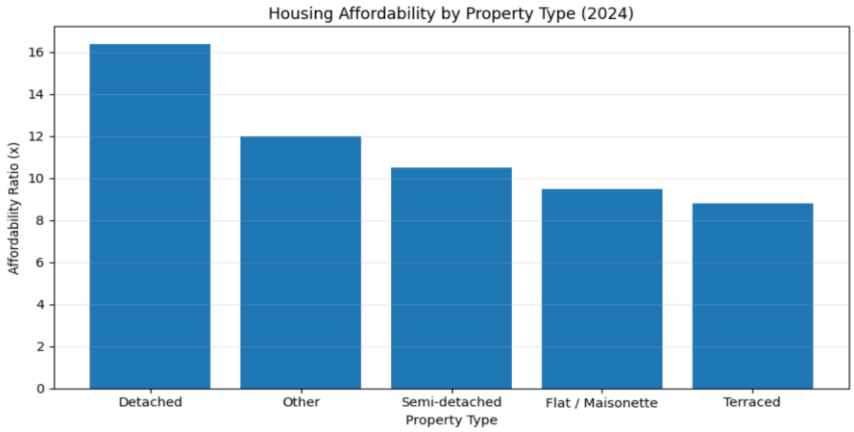
   House prices and local incomes were strongly related across all 318 areas. This relationship became stronger from 2020 to 2024, with Pearson correlation increasing from 0.728 to 0.883 and R^2 rising from 53% to 78%. This means local income explained more of the differences in house prices over time.

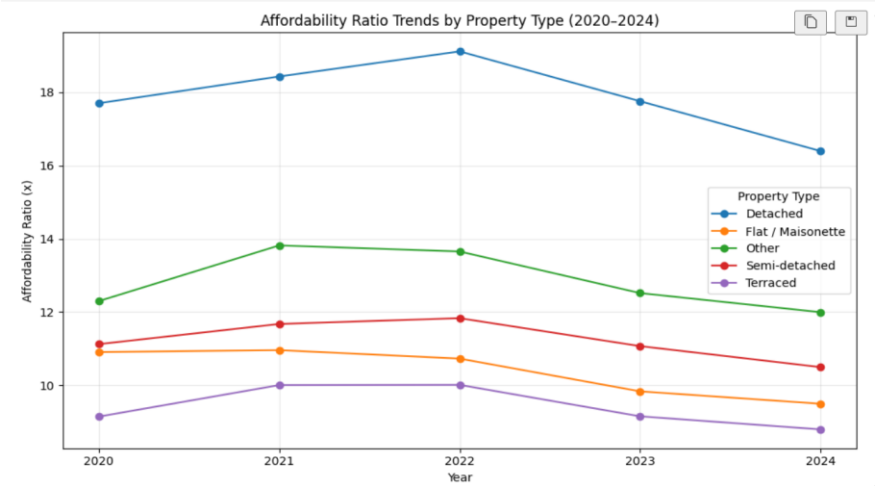
Local Income vs House Price Correlation (2020-2024)

Year	Pearson r	R^2 (Variance Explained)
2020	0.728	0.530
2021	0.742	0.550
2022	0.835	0.697
2023	0.873	0.762
2024	0.883	0.779

4. Does affordability differ by property type?

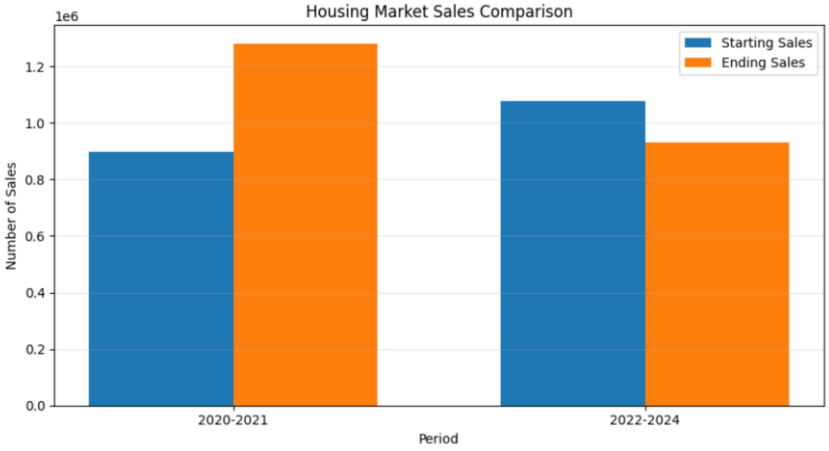
   Affordability varies by property type. In 2024, detached houses were the least affordable, while terraced houses were the most affordable. Detached homes were 86.4% more expensive than terraced homes. During the pandemic, demand for larger homes increased, pushing detached house affordability to a peak in 2022. Flats experienced slower price growth.





5. How did prices respond round 2020 to 2021 compared with 2022 to 2024?

   During the pandemic (2020-2021), there was a 42.75% increase in property sales and an 8% rise in house prices. In contrast, rising interest rates (2022-2024) caused sales to fall by 13.48%. House prices also remained unchanged at £280,000, showing how higher interest rates slowed the housing market.



Limitations

- Income Measurement: GDHI per head does not account for household size, multiple earners or income inequality.
- Missing Property Details: HM Land Registry Price Paid Data lacks property size, number of bedrooms and energy ratings, which can affect house prices.
- Rental Market: The analysis focuses only on house purchases and does not consider rental affordability.

References

1. HM Land Registry (2024). Price Paid Data (2020–2024). Open Government Licence v3.0. UK Government.
2. Office for National Statistics (2024). Gross Disposable Household Income per Head by Local Authority (1997–2024). Open Government Licence v3.0.